BALANCING A BALL ON A BEAM

TESTING PRESENTATION

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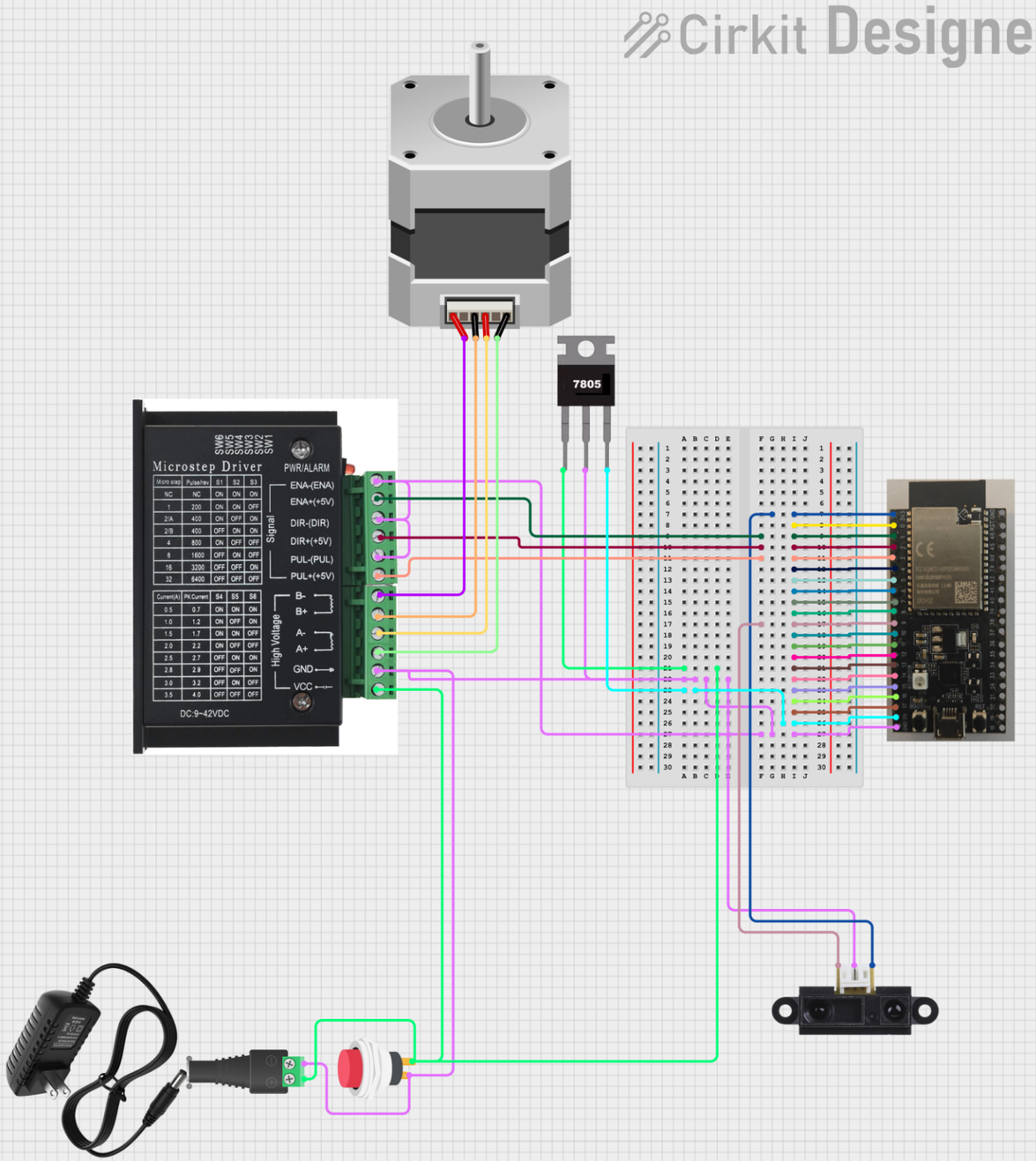
INTRODUCTION

Balancing a ball on a beam using an ESP32 microcontroller is a classic control systems project that demonstrates real-time feedback and actuation

Objective: : To balance a ball on a beam by controlling its position in real-time.

Components: ESP32 S2 Soala 1 microcontroller, TOF sensor(SHARP 2YOD21 F 25), Stepper motor, Power supply 3-12V, USB power supply 4C, Micro step driver, Semi Breadboard, Beam(frame), Ball

Applications: Demonstrates control principles, real-time sensing, and microcontroller programming



Electrical Connections

The electrical connections ensure communication and power flow between the components. The diagram showcases how each component is wired to the others.

TESTING SETUP

ESP32 Microcontroller

The brain of the system, processing sensor data and sending commands to the motor.

Time-of-Flight (ToF) Sensor

Measures the distance between the sensor and the ball, providing feedback for the control system.

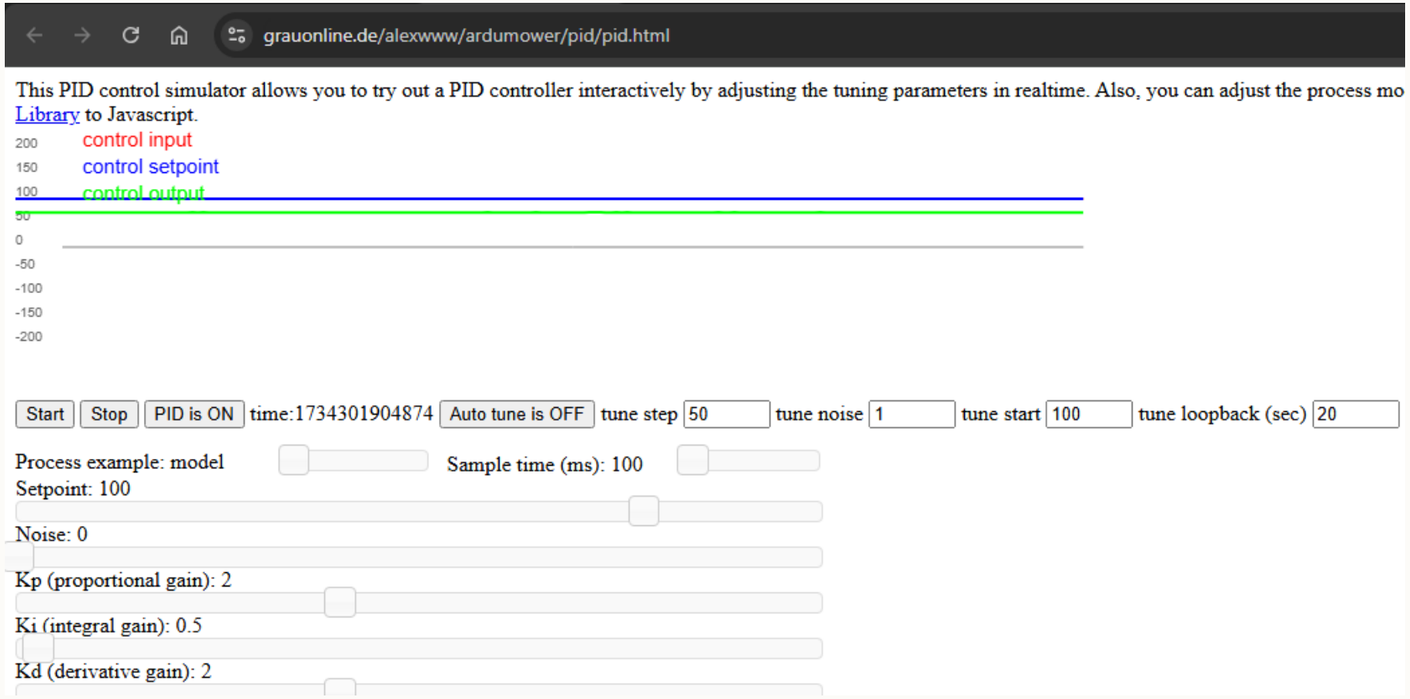
Microstep Driver and Stepper Motor

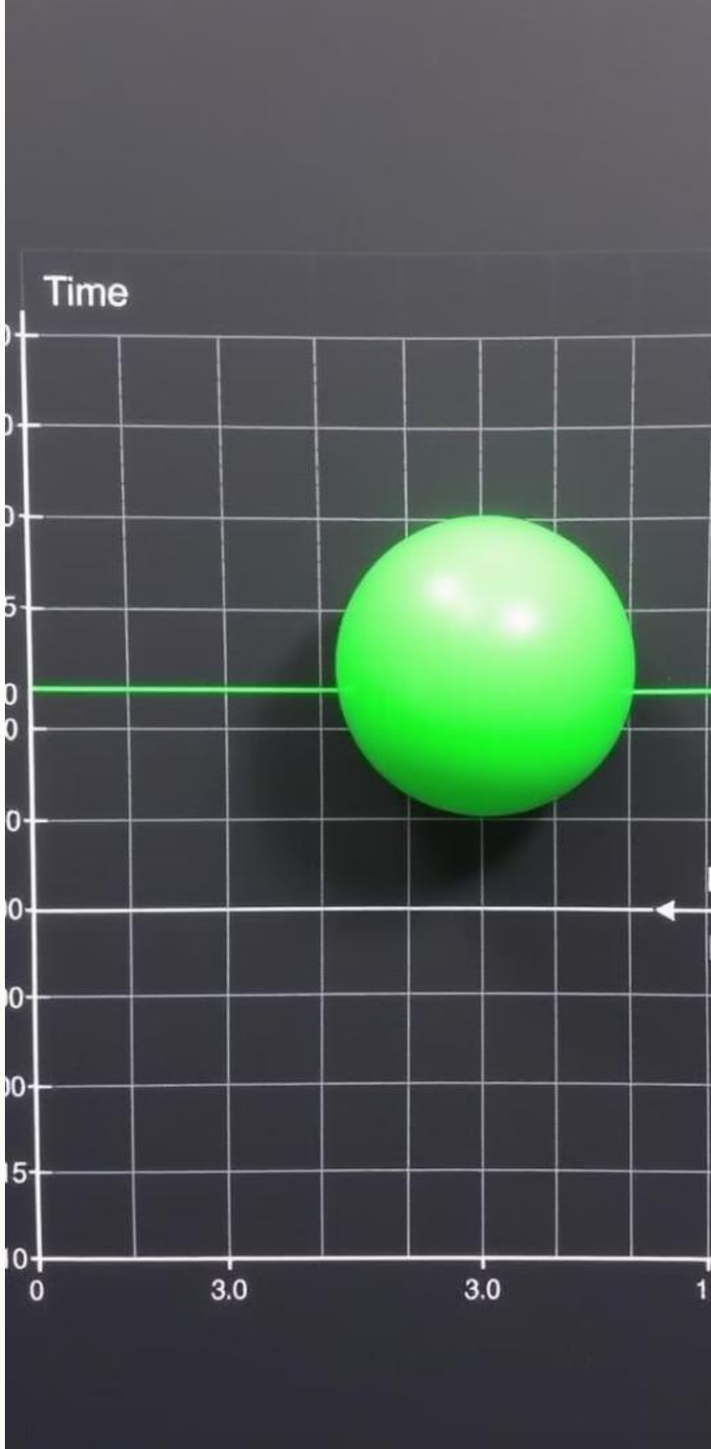
The motor moves the beam, controlled by the ESP32 to keep the ball balanced.

Power Supply

Provides power to all components.

PID Controller Testing





Conclusion

- The electrical connections are all correct
- The Voltage regulator provides a steady and continues voltage of 5V
- Microstepdriver is providing exact predicted pulses into the stepper motor
- ESP32 provides the exact desired output to Microstepdriver
- Stepper motor is working in accordance with the pulse from the Microstepdriver

Thankyou